Builder Journal — Week 1

AI Summer Internship 2026 | Assignment 6

1. Why did you choose AI Engineering?

I chose AI Engineering because I was fascinated by how quickly large language models went from research demos to tools people use every day for real work. What pulled me in specifically was seeing how a handful of well-designed prompts and tool integrations could turn a raw model into something genuinely useful, like a coding assistant or a research agent. Compared to pure ML research, which focuses on training new models, I was more drawn to the engineering side: building systems, APIs, and products on top of models that already exist. This felt like the fastest way to combine my interest in software development with the excitement around modern AI.

2. What type of AI products do you want to build?

I'm most interested in building practical productivity and developer tools — applications that help people work faster and think more clearly, rather than novelty chatbots. Assistants that can read documents, summarize information, and automate repetitive tasks feel especially valuable to me, since they save real time for real people. Long-term, I'd like to work on tools that sit at the intersection of AI and everyday workflows, similar to coding copilots or AI-powered research assistants, where the value is measurable and immediate rather than purely experimental.

3. What was your biggest learning this week?

My biggest learning this week was understanding how an "AI application" is really just a thin layer of engineering — prompts, state management, and error handling — wrapped around an API call to a model. Before this week, I thought building an AI app meant something much more complex, closer to training a model from scratch. Building the AI Workspace app showed me that most of the real engineering effort goes into things like managing conversation history, designing a clean system prompt, and handling failures gracefully, not the AI itself. That completely changed how I think about what "AI Engineering" actually means as a discipline.

4. What challenges did you face?

The first challenge was setting up my development environment correctly — getting the right Python version, creating a virtual environment, and making sure all the packages installed without conflicts took longer than expected. The second challenge was understanding how to manage conversation history in Streamlit, since the app resets every time the script reruns unless state is stored properly. I also found it tricky at first to write error handling that gave genuinely helpful messages instead of just showing a raw error trace to the user.

5. How did you solve them?

For the environment setup, I worked through the installation steps one at a time and used official documentation whenever an error message was unclear, rather than trying to fix everything at once. For the conversation history issue, I learned about Streamlit's session state, which persists data across reruns within the same session, and used it to store both messages and settings. For error handling, I tested the app deliberately with bad inputs — an empty message, a wrong API key, no internet connection — and wrote specific `except` blocks for each case so the user always gets a clear, human-readable message instead of a crash.

6. What are your goals for Week 2?

- Get comfortable with tool calling and connect at least one real external tool (e.g., a search or file tool) to an agent
- Practice designing a basic planning loop so an agent can break a goal into steps on its own
- Commit code to GitHub daily, with clear commit messages, instead of one large commit at the end of the week
- Read at least one foundational paper on agents (e.g., ReAct) and summarize it in my own words